Table of contents

Potential: "intrinsic energy of inertia"

Force caused by Potential: Electro-Magnetic Field

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := rot(A) \quad \overset{\text{Matter}}{=} \mu_0 \left(\mathcal{H} + \mathcal{M}\right) \overset{\text{Vacuum}}{=} \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \quad \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0_{\ldots}}\right. \tag{1}$$

Magnetic Field in Matter

Exitation $\hat{=}$ Displacement

$$\text{Excication:} \qquad \mathcal{H} = \frac{1}{\mu_0}\mathcal{B} - \mathcal{M} = \frac{1}{\mu}\mathcal{B} \quad \text{Magnetisation:} \qquad \mathcal{M} = \frac{\mu - \mu_0}{\mu\mu_0}\mathcal{B} = \frac{\mu - \mu_0}{\mu_0}\mathcal{H} \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t}U_m \quad \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1 - \frac{r}{|r|}\cdot\frac{v}{c}\right)^3} q \left[\underbrace{\frac{\left(1 - \frac{v^2}{c^2}\right)\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \ldots\right.\right.}_{\text{accelerate}\ldots}\right. \tag{3}$$

Electric Field in Matter

Displacement $\hat{=}$ Exitation (field change)
dielectricum = non-conducting material
electricum = conductor medium

Exication: $$\mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1+\chi) \mathcal{E} = \quad \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \tag{4}$$

Abbreviations

Φ = electric potential $\quad A$ = magnetic potential $\quad j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\quad \mathcal{B}$ = Magnetic Field $\quad \nabla \cdot \mathcal{B} = 0 \quad \mathcal{D}$ = Displacement density of dielectricum $\quad \mathcal{M} =$

$\mathcal{P}$ = Polarisation $\quad \mathcal{H}$ = magn. exitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = constant \quad \chi$ = elec. susceptibility of dieelectricum $\quad \kappa = \dfrac{\mu - \mu_0}{\mu_0}$ = magn. susceptibility

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\quad \langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field $\quad \mu$ = magn. permeability $\quad \mu_0$ = magn. constant ...

ε = dielectricity constant $\quad \varepsilon_0$ = ...

Interdependencies of Fields: Electro-Magnetic System (Maxwell conditions)

Static, kinetec rest charge localisation (conservativ system):

...

Stationary, slow charge movement localisation (conservativ system):

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t}\mathcal{B} = i\omega\mathcal{B} \qquad \nabla \times \frac{1}{\mu_0}\mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} = j - i\omega\varepsilon_0\mathcal{E} \tag{5}$$

Dynamic, fast charge movement (relativistic):

...

Potential derivation

$\Rightarrow$ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}U = J \tag{6}$$

$\Rightarrow$ Potential caused by "ignition J" (inhomgenity) from source at rest, v = 0:

$$U = -\frac{1}{4\pi}\int \frac{J}{|r|}\,dr' \tag{7}$$

elec. Potential structure

$J = -\frac{1}{\varepsilon_0}\varrho$

$$U_e := \Phi \overset{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi}\frac{1}{\varepsilon_0}\int \frac{\varrho}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}Q = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}It \tag{8}$$

magn. Potential structure

$J = -\mu_0 j$

$$U_m := A \overset{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|}\, dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\mu_0 I = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{c^2\varepsilon_0} I = \frac{1}{c^2}\frac{1}{t} U_e \overset{\text{source in uniformal movement},v=const}{=} \tag{9}$$

Electric endogen energy

- Bond Energy
- Binding Energy

$E_{pot} = mc^2$

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{10}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{\left(mc^2\right)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \tag{11}$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Exitation Energy

$E_{kin} = \frac{1}{2}mv^2 \mathrel{\hat{=}} \left(pc\right)^2 \mathrel{\hat{=}} h\nu = \hbar\omega$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\varepsilon_0 \mathcal{E}^2 \tag{12}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{13}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{14}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{15}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \tag{16}$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda\frac{1}{T} = \sqrt{\frac{k}{m}} \overset{vacuum}{=} kc \overset{matter}{=} k\frac{c}{\eta} = kc\frac{\tan\beta}{\tan\alpha} \overset{\text{electromagnetic}}{=} kc\left(\mathcal{E}_\beta^{-1}\mathcal{E}_\alpha\right)^{-1} = kc\frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} \tag{17}$$

Phase:

$$\varphi = k\left(\vec{r}\cdot\vec{k}\right) - \omega t := ks - \omega t = ... \tag{18}$$

Phaseshift:

$$\Delta\varphi = \Delta k\left(\vec{r}\cdot\vec{\Delta k}\right) - \Delta\omega t := \Delta ks - \Delta\omega t = ... \tag{19}$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\lambda_0}{\lambda} = c\sqrt{\varepsilon\mu} \mathrel{\hat{=}} \mathcal{E}_\beta^{-1}\mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = ... \tag{20}$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielctric madium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$... \tag{21}$$

Dielectricity:

$$\varepsilon(\omega) = ... \tag{22}$$

Permeability:

$$\mu(\omega) = ... \tag{23}$$

Conductivity:

$$\sigma(\omega) = ... \tag{24}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \tag{25}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{26}$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \tag{27}$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \tag{28}$$

$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{29}$$

$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electic Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emgergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu\lambda = \frac{\lambda}{T}$
$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \qquad \nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index η emerges: $\varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$
Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

static, vacuum, granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

static, vacuum, flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2} = 1$$

Dissipation in Vacuum

no Contiunuity of chrge/matter, no Refraction index, Reflection

Dissipation in Matter

no Contiunuity of charge/matter with dielectricum, refraction merges, reflection, absorbtion

Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

$$\begin{gathered} h = \tan(\varphi) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{gathered} \tag{30}$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \tag{31}$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{32}$$

[1]Wirkungsquantum

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\varphi = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}$$

$$C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}\sqrt{1 - \frac{C}{L}R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2$$

Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{h}{k\lambda}\omega$

Energy released (gained), see… Equation 6

- with mass (e.g. electron), Tunnel-Effect: $E_{gain} = E_{pot} - E_i = mc^2 - h\nu_i$
- massless (e.g. photon), Doppler-Effect (frequency shift): $E_{gain} = E_{kin} - E_i = h\left(\nu_0 - \nu_i\right) = \Delta\omega\hbar$

Electromagnectic Energy Exchange in Vacuum

charge conservation:

$$\frac{\partial}{\partial t}\varrho + \nabla \cdot j = 0 \tag{33}$$

$\varrho = \sigma\delta = \varepsilon_0 \nabla \cdot \mathcal{E} =$ charge density in interior

$\sigma =$ charge density at border (skin)

$\delta =$ Dirac delta-function

$j=\sigma\left(\mathcal{E}+v\times\mathcal{B}\right)=\nabla\times\mathcal{H}-\frac{\partial}{\partial t}\mathcal{D}=\frac{1}{\mu_0}\nabla\times\mathcal{B}-\varepsilon_0\frac{\partial}{\partial t}\mathcal{E}\hat{=}$ elecectric current density

$j\cdot\mathcal{E}\hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S}=\mathcal{E}\times\mathcal{H}\hat{=}$ energy transport density (Intensity $=\left|\overline{\mathcal{S}}\right|$)

Energy conservation for (large extended) conductive medium:

$$\int\left(\frac{\partial}{\partial t}\left(w_e+w_m\right)+\ \cdot\mathcal{S}+j\cdot\mathcal{E}\right)dr=\frac{d}{dt}W_e+\frac{d}{dt}W_m+D_S+\int\left(j\cdot\mathcal{E}\right)dr\overset{\text{low frequency}}{=}\frac{1}{2}\frac{d}{dt}\left(LI^2\right)+\frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right)\tag{34}$$

Energy Balance

Electro-Magnetic Energy:

$$E=E_{kin}+E_{pot}=\sqrt{\left(pc\right)^2+\left(mc^2\right)^2}\overset{\text{vacuum}}{=}pc=\hbar\omega\quad=\text{chargeenergy}+\text{fieldenergy}+\text{transportenergy}\tag{35}$$

Charge Energy:

$$\underbrace{j\cdot\mathcal{E}}_{\text{charge energy}}=-\left(\underbrace{\frac{\partial}{\partial t}\sum w_i}_{\text{field energy}}+\underbrace{\nabla\cdot\mathcal{S}}_{\text{transfer energy}}\right)=\mathcal{E}\cdot\nabla\times\mathcal{A}-\mathcal{E}\cdot\frac{\partial}{\partial t}\mathcal{D}\tag{36}$$

Field Energy:

$$\frac{\partial}{\partial t}w=\frac{\partial}{\partial t}\sum w_i=\frac{\partial}{\partial t}\left(w_e+w_m\right)=\frac{\partial}{\partial t}\left(\frac{\varepsilon_0}{2}\mathcal{E}^2+\frac{1}{2\mu_0}\mathcal{B}^2\right)\overset{vacuum}{=}\frac{\partial}{\partial t}\left(\frac{1}{2}\frac{1}{\mu_0}\left(\mathcal{A}\cdot\nabla\times\mathcal{B}-\mathcal{B}^2\right)\right)\tag{37}$$

Transfer Energy:

$$-\dot{w}\hat{=}\nabla\cdot\mathcal{S}\overset{matter}{=}\nabla\cdot\left(\mathcal{E}\times\mathcal{H}\right)\overset{vacuum}{=}\nabla\cdot\left(\mathcal{E}\times\mathcal{B}\right)\overset{\text{alternativ in vacuum}}{=}\nabla\cdot\left(\frac{1}{2}\frac{1}{\mu_0}\frac{\partial}{\partial t}\left(\mathcal{A}\times\mathcal{B}\right)\right)=...\tag{38}$$

Local Energy Conservation:

Force $F=q\left(\mathcal{E}+v\times\mathcal{B}\right)$

Work $W=\Delta E=Fr=pv=const.$

Power $\frac{d}{dt}W=0$

$$\overbrace{\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}}}^{= \text{const.}} + \underbrace{\cdot \sum \mathcal{S}_{w_i}}_{\text{transfer energy}} \overset{alternatives}{=} j \cdot \mathcal{E} + \frac{\partial}{\partial t} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{A} \cdot \nabla \times \mathcal{B} \right) + \frac{1}{\mu_0} \nabla \cdot \left(\mathcal{E} \times \mathcal{B} + \frac{1}{2} \frac{\partial}{\partial t} \right. \tag{39}$$

conditions for modeling alternatives (Eigenvalue solution):

- $w = \sum w_i : \frac{\partial}{\partial t} w = -\nabla \cdot \mathcal{S}_w$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$

Energy Density Flux: Pyointing-Vector

$$\mathcal{S} \overset{vacuum}{=} \mathcal{E} \times \mathcal{B} \overset{matter}{=} \mathcal{E} \times \mathcal{H}$$

Far Field: Power $= \frac{\Delta E}{t} \hat{=} \frac{d}{dt} E = 0 \Rightarrow$ Energy $=$ const. $\Rightarrow$ Energy Conservation

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}}}_{= \text{const.}} \right) = 0$$

Near Field (local): Power $=$ Energytransport

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) \overset{local}{=} \underbrace{- \int_{\mathbb{R}^2} \mathcal{S} \cdot \mathbf{n} \underbrace{df}_{\text{"skin"}}}_{\text{transported energy density}}$$

Local Force Balance

Electro-Magnetic Force:

$$\dot{p} = F = q\left(\mathcal{E} + v \times \mathcal{B}\right) \quad = \text{chargeforce} + \text{fieldforce} + \text{stresstension} \tag{40}$$

Charge Force:

$$... = \varrho\mathcal{E} + j \times \mathcal{B} \tag{41}$$

Field Force:

$$... = \frac{\partial}{\partial t}\frac{1}{c^2}\mathcal{S} \tag{42}$$

Tension Force:

$$\nabla \cdot \mathcal{T} = \varepsilon_0\left(\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E}\nabla \cdot \mathcal{E}\right) + \frac{1}{\mu_0}\left(\mathcal{B}\nabla \times \mathcal{B} - \mathcal{B}\nabla \cdot \mathcal{B}\right) \tag{43}$$

Local Momentum Conservation:

Force $F = q\left(\mathcal{E} + v \times \mathcal{B}\right) = \frac{d}{dt}p = 0$

Momentum $p = \int F = \frac{E}{c} = const.$

$$\overbrace{\underbrace{\varrho\mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t}\frac{1}{c^2}}_{\text{field force}}\mathcal{S}}^{...} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}} = 0 \tag{44}$$

Tension Force: Stress-Tensor

Stress Tensor quantifies tension $\widehat{=}\Delta$ Forces = Pressure - Traction $\widehat{=}$ "torsion tension" at angular momentum

$$\mathcal{T} \widehat{=} \mathcal{T}_{ik} = \delta_{ik}\left(\frac{1}{2}\varepsilon_0\mathcal{E}^2 + \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2\right) - \varepsilon_0\mathcal{E}_i\mathcal{E}_k - \frac{1}{\mu_0}\mathcal{B}_i\mathcal{B}_k \tag{45}$$

$\mathcal{E}$ and $\mathcal{B}$ fields of spacial confined charge and current density distribution, far field, static: $\mathcal{T} \sim \frac{1}{r^4} \Leftrightarrow \int_{\mathbb{R}^3} \mathcal{T} d^3r = \int_{\infty} \mathcal{T} df = 0$

Far Field: Force = 0 $\Leftrightarrow$ Momentum = const. $\Rightarrow$ Momentum Conservation

$$F = \dot{p} \,\widehat{=}\, \frac{d}{dt}\underbrace{\left[\underbrace{\int_{local}\left(\varrho\mathcal{E} + j \times \mathcal{B}\right)}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2}}_{\text{field momentum}}\mathcal{S}\right]}_{=p=const.} = 0$$

Near Field (local): Force = Tension

$$F = \dot{p} \mathrel{\widehat{=}} \frac{d}{dt}\underbrace{\left[\underbrace{\int\limits_{local} (\varrho\mathcal{E} + j\times\mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{\underbrace{c^2}}\mathcal{S}}_{\text{field momentum}}\right]}_{=p\neq const.,\ \exists -\mathcal{T}\ \text{local tension}} \overset{local}{=} \underbrace{-\int\limits_{\mathbb{R}^2} \mathcal{T}\cdot\mathbf{n}\ \underbrace{df}_{\text{"skin"}}}_{\text{stress tension appears}}$$

Momentum Balance

Momentum Conservation and Local Tension: linear momentum conservation if $p = const.$, otherwise "tension" occures:

$$p = \int F = \int \frac{d}{dt}p = \sqrt{\frac{1}{c^2}\left(E^2 - (mc)^2\right)} \overset{\text{vacuum}}{=} \frac{E}{c} = \frac{2\pi}{c^2}h\nu$$

$p = const.$, linear, vacuum, Far Field:

$$F = \dot{p}\mathrel{\widehat{=}}\int\limits_{\mathbb{R}^3}\frac{d}{dt}\left[\underbrace{(\varrho\mathcal{E} + j\times\mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t}\frac{1}{c^2}\mathcal{S}}_{\text{field force density}}\right] = \frac{d}{dt}\underbrace{\left[\underbrace{\int\limits_{\mathbb{R}^3}(\varrho\mathcal{E} + j\times\mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{\underbrace{c^2}}\mathcal{S}}_{\text{field momentum}}\right]}_{=\text{ p momentum conserved if p = const.}} = 0 \tag{46}$$

$p \neq const.$, local $\in \mathbb{R}^2 \ll \mathbb{R}^3 = \infty$, linear, vacuum, Near Field $\Leftrightarrow \exists -\mathcal{T}$ as tension compensation flux:

$$F = \dot{p}\mathrel{\widehat{=}}\int\limits_{local}\frac{d}{dt}\left[\underbrace{(\varrho\mathcal{E} + j\times\mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t}\frac{1}{c^2}\mathcal{S}}_{\text{field force density}}\right] = \frac{d}{dt}\underbrace{\left[\underbrace{\int\limits_{local}(\varrho\mathcal{E} + j\times\mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{\underbrace{c^2}}\mathcal{S}}_{\text{field momentum}}\right]}_{=p\neq const.,\ \exists -\mathcal{T}\ \text{local tension}} \overset{local}{=} -\int\limits_{\mathbb{R}^2}\mathcal{T}\cdot\mathbf{n}\ \underbrace{df}_{\text{"skin"}} \tag{47}$$

Momentum $p = \int\limits_{\mathbb{R}^3}(\varrho\mathcal{E} + j\times\mathcal{B}) + \frac{1}{c^2}\mathcal{S}$

photon Spin $= \int\limits_{\mathbb{R}^3}\frac{1}{c^2}\mathcal{S}$

angular momentum (photon Spin):

$$D = \dot{L} \widehat{=} ... \tag{48}$$

Radiation Pressure

average energy density $\widehat{=}$ radiation pressure, if stationary (very slow flow motion $v \ll c$) :

$$\overline{w} \widehat{=} -\nabla \mathcal{T} \cdot \mathbf{n} \widehat{=} -\frac{1}{c} \left|\overline{\mathcal{E} \times \mathcal{H}}\right| = -\frac{1}{c} \left|\overline{\mathcal{S}}\right| = \overline{(\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}} \overset{stationary}{=} \overline{\varrho \mathcal{E} + j \times \mathcal{B}} \tag{49}$$

Abbreviations:

w_i = energy densities (electric w_e, magnetic e_m, gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectros $\widehat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Stress Tensor, (tension, momentum flow density)

$\mathbf{n}$ = surface normal vector (orthongoal on skin)

$\Delta \cdot :=$ div grad = Laplacian

J = "ignition of inhomogenity,cause"

j = Current density

ϱ = Charge density

V = Potentialdifference

I = Current

R = Resistance

L = Inductance

C = Capacitance

Q = charge (ion)

$\Psi = LI$ = magn. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_S = \int (\cdot \mathcal{S})\, dr = \int \mathrm{div}(\mathcal{E} \times \mathcal{H})\, dr$ = dissipation engergy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}}$ = skin depth (surface) of conductor

$r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho$ = radius (extension) of conductor

$\rho = kr$ = "radius of curvature k"

μ = permeability of conductor

σ = conductivity of conductor (dielectricum)

Electromagnectic Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Vacuum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Vaccumm

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Vacuum

. - Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast) - Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Localization and Fields of Energy Transport

Localization of energy density w and of its energy transport $\mathcal{S}_w$ is where field force has strongest density. Fields for Forces, Energy, Momentum, Matter:

- Electromangnetic:
$$\left.\begin{array}{l}\text{Electric: } \mathcal{E}, \mathcal{D}, \mathcal{P}, \Phi_e \\ \text{Magnetic: } \mathcal{B}, \mathcal{H}, \mathcal{M}, \mathcal{A}, \Phi_m\end{array}\right\} \mathcal{S}, \mathcal{T}$$
- Thermal (skalar): T
- Gravitational: G

Frequency Cause-Effect

Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta k v = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \overset{electromagnetic}{=} ... = \sqrt{\frac{1}{LC}} \overset{QM}{=} \frac{1}{2}\frac{p^2}{m}\frac{1}{\hbar} \tag{50}$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v}\frac{1}{\lambda} = \frac{\omega}{k}\frac{1}{\lambda} = \frac{\omega}{2\pi} \tag{51}$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{\hbar}\omega\mathcal{T} + \nu_0 \tag{52}$$

$$\begin{array}{r}e = \text{electron charge} \qquad V_0 = \text{Opposing Potential } (I = 0) \\ \text{Energy Level: } i = [1, n] \in \mathbb{N} \qquad\qquad\qquad\qquad\qquad\qquad\qquad\end{array} \tag{53}$$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \tag{54}$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurance of an event/state by space distance and time periodicity as well as movement speed and reach length, $\quad v^2 = v_{ph}v_{gr}, \quad \omega t = 2\pi\nu t = \lambda \neq 0 \quad$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \tag{55}$$

Rest state of source charge

Statics localization of charge. Only spacial demependeny, no time dependency.

Energy transport ... Information transmission ...

$v = 0 \qquad \dot{v} = 0$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v = const. \ll c \qquad \dot{v} = 0$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \approx c \qquad \dot{v} = 0$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \neq 0 \qquad \dot{v} \neq 0$

Movement in Vacuum

$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{c}{v_{Ph}} = 1 \qquad 2\pi = k\lambda \qquad v_{ph} = c \qquad \omega = kc$

$\varepsilon_0 \mu_0 = \frac{1}{c^2}$

Continuity: $\nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$

Movement in Matter

$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{c}{v_{Ph}} = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \neq 1$ $\qquad v_{ph} = \pm\sqrt{\frac{1}{\varepsilon\mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}}$ $\qquad \omega = kv_{ph} = kc\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} = kc\frac{\tan\beta}{\tan\alpha} = k\frac{c}{\eta}$

$\varepsilon\mu = \left(\frac{\eta}{c}\right)^2$

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$r_0 \ll r \ll \lambda \qquad kr \ll 2\pi \qquad \frac{w_m}{w_e} \ll 1$

r is small, fast movement

Far Field

$r \gg \lambda \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1 \qquad \frac{w_m}{w_e} = 1$

r is big, slow movement

High Frequency

$\omega \gg \frac{2}{\mu\sigma r_0^2}$

λ is big, ν frequency i small, short time period

Low Frequency

$\omega \ll \frac{2}{\mu\sigma r_0^2}$

λ is small, ν frequency i big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global: ....

Local: ....

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1}\mathcal{D} = \mathcal{E}^{-1}\left(\mathcal{P} + \varepsilon_0\mathcal{B}\right)$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1}\mathcal{H} = \mathcal{B}^{-1}\left(\frac{1}{\mu_0}\mathcal{B} - \mathcal{M}\right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1+\omega^2 t^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta\frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

Conditions (skin, interface, border, surface)

Boundary Conditions

$\mathbf{n} \cdot [\mathcal{D}] = \sigma$

$\mathbf{n} \cdot [\mathcal{B}] = 0$

$\mathbf{n} \times [\mathcal{E}] = 0$

$\mathbf{n} \times [\mathcal{H}] = J$

Interior Conditions

e.g. for Superconductivity

$\mathbf{n} \cdot \mathcal{H} = 0$

$\mathbf{n} \times \mathcal{D} = 0$

Abreviations

$\mathbf{n}$ = normal vector (orthogonality orientation)

$\mathcal{E}$ = elec. Field (“traction force”?)

$\mathcal{B}$ = magn. Field (“compressive force”?)

$\mathcal{D}$ = elec. Displacement Field (“exitioan”)

$\mathcal{H}$ = magn. Exitation Field (“incentivation”)

σ = electric charge at surface

J = current density at surface

Distinguishing Situations Conditions and Dependencies

effects of charge (separaton) and differentieting situations (conditioning frameworks) for fields and potentials:

Spacetime Embeding

- Spacial dependence (mechanics)
- Time dependence (dynamics)

Movement Framework

- Static: rest system, equilibrium
- Dynamic:
  - Stationary: slow motion
  - Relativistic: fast motion
  - Radiation; accelerated motion

Medium Property

- Vacuum: spacial mean = time mean
- Matter:
  - Mass (inertia)
  - Charge (conductor, ion)
  - Dielectricum (isolator)
  - Dispersion (refraction)

Field Distance

- Near Field (granular spacetime, mean average values)
- Far Field (flat spacetime)

Localisation Identity

- Global (regular)
- Local (singular)

-> Tool concepts: Momentum, Force, Energy

-> Conditions framework: - continuity condition (charge, current) - equilibrium conditions (action vs. reactio, homgenity vs. inhomogenity) - initial conditions (starting point) - border (skin) conditions (differentiation) - interior conditions (integration) - conservations of local densities (charge, energy, momentum)

-> Properties: - characteristics from charge: dielectricity ε, permeability μ, conductivity σ, frequency ω - characteristics from heat: … - Properties frm gravitation: …

Operators

$\nabla \coloneqq grad =$ Gradient (Gefälle, Neigung)

$\nabla \cdot \coloneqq div =$ Divergence (Quelle)

$\nabla \times \coloneqq rot =$ Rotation (Wirbelung, Curl)

$\nabla^2 =$ Lagrange Operatonr (Krümmung, Beschleunigung)

$\Delta \cdot \coloneqq div(grad) =$ Laplace Operatorn

$\Delta \coloneqq$ Difference, shift